

Volume 41

Sentinel Endpoint Security Engine (SESE)

Status: Core Security Domain

Priority: Critical

Vision

Every computer should have a continuously updated digital health profile.

Not only:

- Is malware detected?

Instead:

- Is the device healthy?
- Is it trusted?
- Has anything changed?
- Is someone attempting persistence?
- Is configuration drifting?
- Is recovery possible?
- Can the user understand everything?

The endpoint becomes a Living Object.

Supported Platforms

The architecture is designed for:

- Windows
- Linux
- macOS

using platform-specific collectors behind one common Sentinel API.

The user should never notice platform differences.

Endpoint Object

Every endpoint becomes a Sentinel Object.

Organization

Everything is connected.

Endpoint Identity

Each endpoint receives a permanent identity.

Example

PS-ENDPOINT-00482931

Identity never depends on the hostname alone.

Endpoint Lifecycle

Manufactured

Every transition becomes an event.

Endpoint Health Model

Rather than one percentage.

Health consists of multiple dimensions.

Examples

Operating System

Security Configuration

Patch Status

Disk Encryption

Firewall

Identity Protection

Application Health

Recovery Readiness

Hardware Status

Privacy

Evidence Freshness

Trust

This creates a richer picture.

Baseline Engine

One of the most important engines.

When an endpoint is healthy.

Project Sentinel creates a verified baseline.

Future observations compare against it.

Example

Healthy State

No guessing.

Continuous Observation

Instead of heavy scans.

The endpoint continuously observes important state changes.

Examples

- New software installed
- Service created
- Driver changed
- Certificate added
- Local administrator modified
- Scheduled task created
- Startup entry changed
- Firewall modified
- Encryption disabled
- Time synchronization issues

Changes become evidence.

Endpoint Evidence

Examples

Operating System Version

Installed Updates

Disk Encryption

Firewall Configuration

Security Software Status

Boot Configuration

Secure Boot Status

TPM Status (where available)

Installed Applications

Startup Programs

Services

Drivers

Scheduled Tasks

Certificates

Network Configuration

Local Accounts

Privilege Assignments

Each evidence item is versioned.

Configuration Drift

One of our strongest enterprise features.

The platform continuously answers:

What changed?

Not simply:

What exists?

Example

Yesterday

Endpoint Timeline

Every endpoint has a complete history.

Example

Purchased

Perfect for investigations.

Trust Calculation

Trust is not based on one observation.

It considers:

Evidence Freshness

Baseline Integrity

Security Controls

Verification History

Configuration Stability

Policy Compliance

Relationship Confidence

No hidden scoring.

User Experience

For private users.

The dashboard should answer only three questions.

Am I Safe?

What Changed?

What Should I Do?

Everything else is available if they want to explore.

Enterprise Dashboard

Security teams see:

Fleet Health

Trust Trends

Configuration Drift

Patch Coverage

Encryption Coverage

Verification Status

Mission Progress

No unnecessary complexity.

NEW CORE ENGINE

Endpoint Integrity Engine

Continuously validates:

- Operating system integrity indicators
- Boot configuration
- Security settings
- System consistency
- Critical configuration changes

The engine reports observations and confidence rather than making unsupported claims.

NEW CORE ENGINE

Endpoint Recovery Engine

One of the features I believe enterprises and home users will value most.

It helps verify recovery readiness by checking items such as:

- Restore points (where supported)
- Backup configuration
- Recovery keys
- Boot recovery options
- System repair capabilities

The emphasis is on preparedness before incidents occur.

NEW CORE ENGINE

Endpoint Behavior Engine

Instead of relying only on signatures.

The engine builds a behavioral baseline.

Examples

Normal Login Times

Normal CPU Pattern

Normal Memory Pattern

Normal Network Pattern

Normal Services

Normal Startup Configuration

When significant deviations occur, the platform records evidence for review. Organizations can configure thresholds and workflows appropriate to their environments.

NEW CORE ENGINE

Endpoint Isolation Engine

For enterprise environments.

When organizational policy permits, workflows can help administrators:

- Restrict network communication.
- Pause selected scheduled activities.
- Collect additional forensic evidence.
- Notify responsible teams.

All actions require appropriate authorization, are fully audited, and are designed to support incident response procedures.

Mobile Relationship

Since mobile devices are increasingly central to people's digital lives, the endpoint engine links seamlessly with the Mobile Security Engine.

Example

Employee

One digital identity.

Multiple devices.

Digital Twin Integration

Every endpoint automatically updates:

- Object Engine
- Knowledge Graph
- Trust Engine
- Event Bus
- Mission Engine
- Compliance Engine
- Dashboards

The Digital Twin always reflects verified changes.

NEW IDEA

Device Passport

Every computer receives its own passport.

Example

Laptop

Simple.

Powerful.

Understandable.

NEW IDEA

Personal Cyber Coach

One of the features that aligns perfectly with your original vision.

Instead of displaying:

Critical Configuration Issue

The platform says:

Your laptop's disk encryption is turned off. Enabling it will better protect your files if the device is lost or stolen. Estimated setup time: about 10 minutes. We can guide you through

the process.

Technical findings become actionable guidance.

NEW IDEA

Fleet Health Map

Large organizations need a visual overview.

Example

Global

Executives see trends.

Administrators can drill down to individual devices.

PS-ADR-0036

Continuous Endpoint Assurance

Status: Accepted

Decision

The Endpoint Security Engine shall continuously collect evidence, detect meaningful configuration changes, maintain verified baselines, and provide transparent recommendations without relying on opaque scoring or unexplained conclusions.

Rationale

Endpoints change constantly. Continuous, evidence-based observation provides more reliable operational awareness than periodic point-in-time scans alone.

NEW LONG-TERM VISION

The Living Endpoint

I believe this should become one of the defining ideas of Project Sentinel.

A laptop is **not** just hardware.

It has:

- A history.
- An owner.
- A trust journey.
- Relationships.
- Policies.
- Evidence.
- Missions.
- Recovery readiness.
- Security posture.

Project Sentinel should treat every endpoint as a living part of the organization's Digital Twin.

A new differentiator

I would like to introduce the **Sentinel Explain Engine**.

Every recommendation—whether for a home user or a SOC analyst—should answer five questions:

1. **What changed?**
2. **Why does it matter?**
3. **How confident are we?**
4. **What evidence supports this conclusion?**
5. **What are the recommended next steps?**

This aligns with our goals of being **easy to use, WCAG-compliant, multilingual, explainable, and suitable for everyone from home users to banks and government agencies**.

Looking Ahead

The next volume will be **Volume 42 - Sentinel Mobile Security Engine (SMSE)**.

Given your emphasis from the beginning that smartphones have become people's primary computing devices, I think this will be one of the flagship modules of Project Sentinel. It will define a unified architecture for Android and iOS, focusing on evidence-based security assessment, device health, privacy posture, application risk analysis, enterprise fleet management, and user-friendly guidance—all within the permissions and security models of each mobile operating system. This will make mobile protection a first-class capability rather than an add-on, consistent with the architecture we've built across the entire platform.

====[USER]=====

Good and continue

====[ASSISTANT]=====

Excellent. I actually think this will become one of the flagship modules of Project Sentinel. You identified this very early in the project:

"Mobile phones are now more important than PCs for many people."

I completely agree.

Most cybersecurity platforms still treat mobile security as an add-on.

Project Sentinel will treat it as a **first-class citizen**.
That will be one of our major differentiators.

PROJECT SENTINEL BLUEPRINT

Volume 42